



01 Executive Summary

SwarmOps solves the critical bottleneck in drone fleet operations: deciding, in real time, which drone flies which mission, in what order and along which route - while managing battery, range and priority constraints as they shift. It pairs a VRPTW optimization engine with a distributed architecture that re-plans the fleet within seconds of a disruption in the field. The prototype works today: deployed on AWS, ready for live demonstration.

02 The Challenge

- **Operational complexity, disconnected tools:** manufacturer (OEM) software flies a single aircraft superbly - and stops there. The tools don't talk to each other, and nothing assembles a fleet-level picture: who is flying what, where every battery stands, and whether each aircraft can actually complete its mission safely.
- **The scale problem starts at two:** one drone is a task; two or more is a coordination problem - who covers what, who already swept it, whose report is current. Past a handful, battery, payload weight, time windows and airspace constraints interact faster than any dispatcher can track, and manual coordination collapses outright.
- **The cost in the field:** wasted flight time, duplicate coverage, needless airframe wear - and worst of all, an operational picture nobody can fully trust.

03 The Solution & Product

VENDOR-NEUTRAL LAYER

A vendor-neutral coordination and optimization layer

- **Assignment & routing engine (VRPTW):** computes optimal routes, allocates missions automatically, and folds charging or battery-swap stops into the plan in real time - maximum air time, minimum risk.
- **Dynamic re-planning:** live telemetry and field events update the mission board within seconds of a technical fault, an abnormal battery drain, loss of signal, or an urgent shift in operational priorities.
- **Unified control room:** centralized fleet control, a live observation feed for commanders and managers, and dashboards tracking mission status, aircraft readiness and remaining air time - one coherent operational picture.
- **Hardware agnostic:** connects seamlessly to mixed fleets from any manufacturer or protocol (e.g. MAVLink), so an organization gets full use of the aircraft it already owns.
- **Deploys where the customer operates:** in the cloud (AWS and equivalents), on local servers in closed networks (on-premise), or on edge nodes in the field - containerized (Docker & Kubernetes), with automated CI/CD pipelines and full observability.

06 Team

Tony Verin (team lead) | Guy | Harel - three engineers and combat veterans, **two of them drone operators**, who designed, built and operate the entire system end to end: algorithms, cloud architecture, user interfaces and infrastructure.

Why We Built This

This is not a theoretical scenario for us. On 7 October 2023, during the fighting in the kibbutzim, two teams from the same army company launched drones at the same time. With no coordination between them, both swept the same sector - one team identified hostiles, the other reported a friendly force in the same place. The contradiction in the situational picture delayed a critical operational decision. **That is the problem SwarmOps was built to solve.**

07 The Ask

To take SwarmOps from working prototype (PoC) to an operational, commercial product, we are looking for:

- A place in a **technology incubator / accelerator**, with hands-on business mentorship.
- **Strategic introductions:** institutional operators, systems integrators and infrastructure companies.
- **Pre-Seed support:** a drone fleet for live testing, cloud resources, and regulatory and legal counsel.

04 Applications & Market Potential

DUAL-USE

Defense, emergency & C4I	One unified operational picture, coordinated sweeps and rapid response; intelligent fleet logistics (battery and airframe health); an ML layer for after-action review, pattern recognition and sector-specific optimization.
Construction & infrastructure (ConTech)	Autonomous mapping keyed to project milestones: earthwork volume calculation, progress verification for releasing lender funds, as-built comparison against engineering plans (BIM), and defensible documentation for disputes.
Infrastructure & energy	Recurring monitoring, defect detection and preventive maintenance across power lines, oil & gas pipelines, national assets and solar farms.
Precision agriculture	Coordinated multispectral surveys for pest detection, and scheduled autonomous spraying at scale.
Logistics & delivery	Autonomous site-to-site transport and last-mile optimization across the supply chain.

05 Roadmap - 12 Months

Months 0-3

Connect real aircraft (MAVLink), run first field trials, begin engagement with the Civil Aviation Authority.

Months 3-6

First pilot with a strategic partner or design customer, proving out agreed success metrics.

Months 6-12

Multi-Tenant product, commercial pricing, and a Pre-Seed / Seed round.